SELLER PERFORMANCE & CUSTOMER EXPERIENCE ANALYSIS IN E-COMMERCE

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PROJECT MOTIVATION & RESEARCH QUESTION

- I am interested in a career as a Data Analyst or Business Analyst for a business, potentially in the retail industry.
- Working with a real business dataset with seller information (about customers, sellers, products, sales, operations, etc.) could give valuable experience.
- E-commerce provides an opportunity to examine business performance and customer experience, using real information
- My project will focus on business performance and customer experience using the dataset, as both are important factors in business decision making.
- Final research question: Which operational, product, and geographic factors are most strongly associated with seller performance and customer satisfaction in the Olist Brazilian e-commerce marketplace?

DATA SOURCE

- Data: Brazilian E-Commerce Public Dataset by Olist
- Can be access through Kaggle
  - https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce
- A Brazilian ecommerce public dataset of orders made at Olist Store.
- Olist connects merchants and their products to the main marketplaces of Brazil.
  - Merchants list their products once through Olist and Olist helps sell those products in multiple marketplaces
- The dataset has information of 100k orders from 2016 to 2018 made at multiple marketplaces in Brazil.
- It is real commercial data, it has been anonymised, and references to the companies and partners in the review text have been replaced with the names of Game of Thrones great houses.

METHODS AND ANALYTICAL APPROACH

CUSTOMER EXPERIENCE

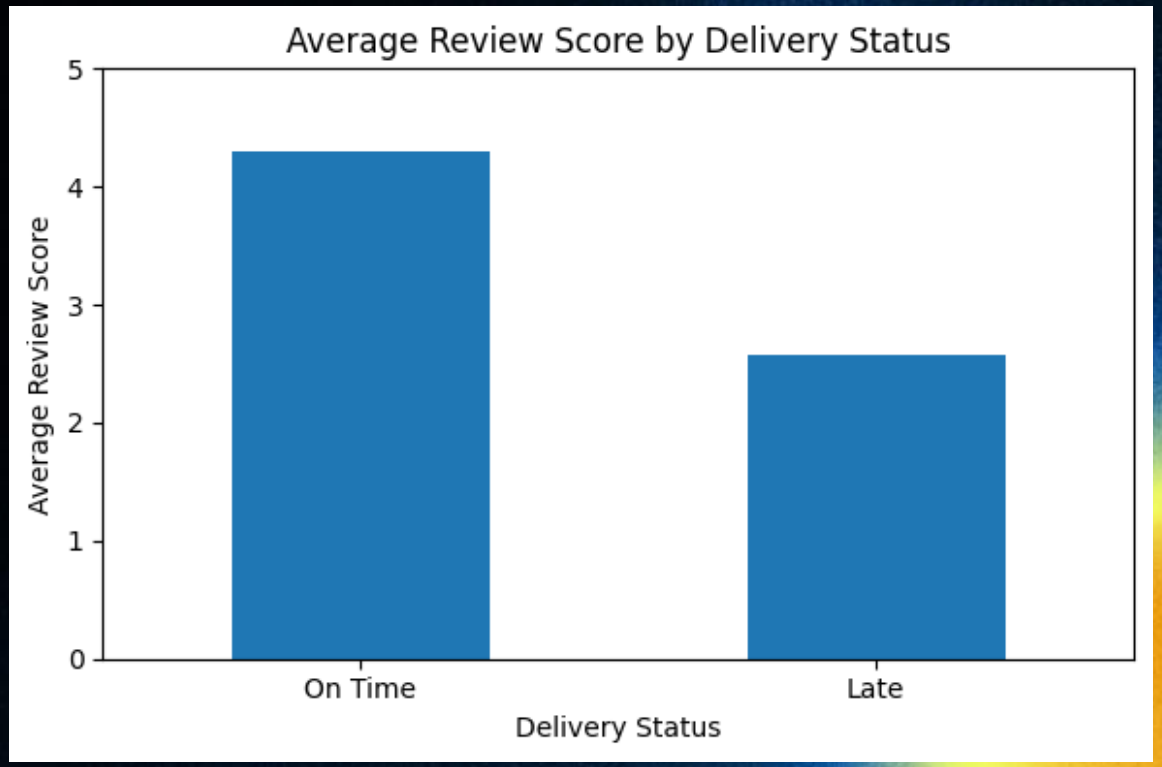
- Main measure: `review_score`
- Explored delivery, order value, freight, product category, seller characteristics, and geography
- Mann-Whitney U:- On-time vs. late reviews
- Spearman correlation:- Delivery delay vs. review score
- Rank-biserial correlation:- Effect size
- Category/state subgroup checks

SELLER EXPERIENCE

- Main measure: `seller_total_sales`
- Examined:- Order count
  - Product count
  - Category count
  - Average item value
- Spearman correlations
- Sensitivity analysis:
  - All sellers
  - Sellers with ≥10 orders

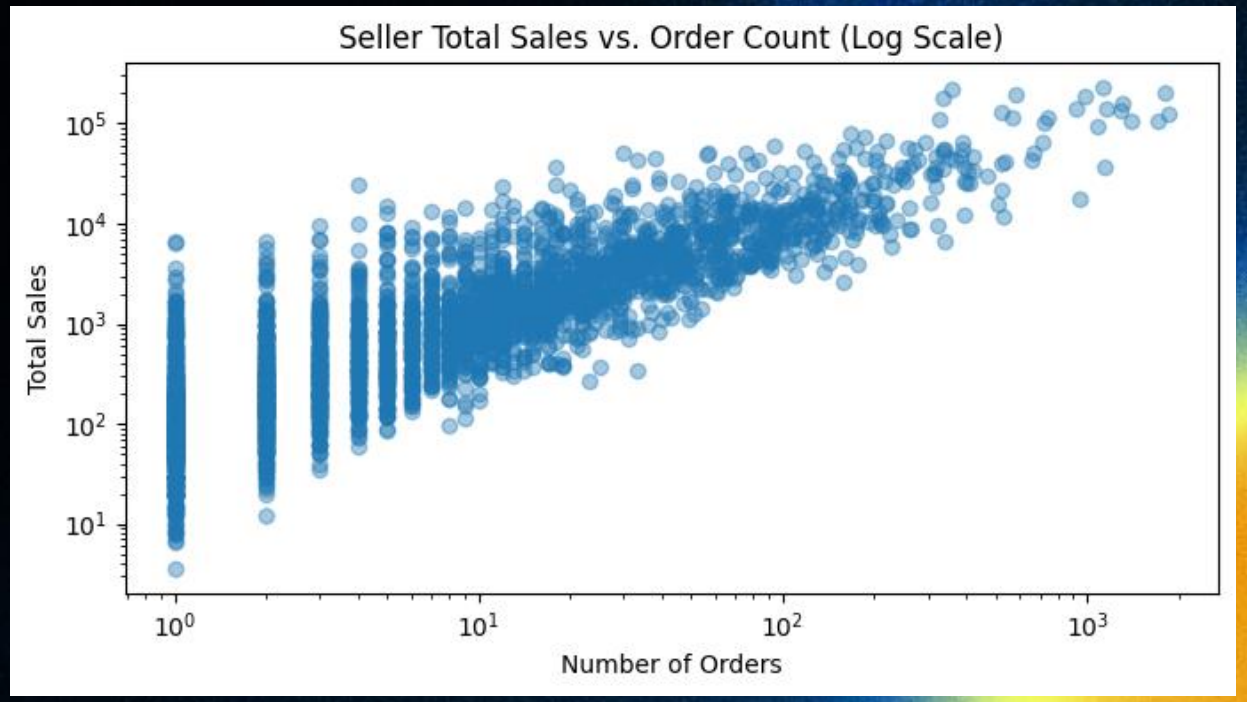
BASELINE/PRELIMINARY RESULTS: CUSTOMER EXPERIENCE

- Delivery performance showed the clearest relationship with review score after exploring multiple factors.
- On-time orders averaged 4.29 vs. 2.57 for late orders.
- Average difference: 1.72 review points.
- Rank-biserial effect size: 0.554.
- Delivery delay had a weaker negative relationship with review score (ρ = -0.176).
- On-time reviews were higher in all 19 qualifying product categories and 14 qualifying customer states.



BASELINE/PRELIMINARY RESULTS: SELLER PERFORMANCE

- Seller activity and sales were highly right-skewed.
- Median seller orders: 6; maximum: 1,854.
- Median seller sales: about 821 BRL.
- Total sales were most strongly related to order count and product count.
- Among sellers with at least 10 orders, order count remained strongly related to sales (ρ = 0.753).
- Product and category-count relationships weakened in the sensitivity analysis.



CHALLENGES SO FAR

- Relational tables required careful joins to avoid duplicated or inflated results.
- Many sellers have very few transactions.
- Seller activity and monetary measures are strongly skewed.
- Seller geography is concentrated in São Paulo.
- Large sample sizes make it important to consider effect size, not only p-values.
- Current findings show association, not causation.

NEXT STEPS

- Incorporate presentation and TA feedback.
- Finalize variables for improved analysis.
- Apply a multivariable statistical approach.
- Compare improved results with the baseline.
- Validate the strongest findings.
- Develop business-focused conclusions and recommendations.
- Do further research to help my findings.
- Start structuring my final report.